

# Arith Maldonado Zamudio

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GPA: 9.1/10.0 | Engineering portfolio: [aritmaldzamu.github.io/portfolio-ingenieria-mecanica/](https://aritmaldzamu.github.io/portfolio-ingenieria-mecanica/)

## Professional Summary

Mechatronics Engineering student and Biomedical Engineer enthusiastic about automation, production, mechanical design, and R&D within advanced manufacturing environments. Hands-on experience in medical equipment service, FEA, embedded prototyping, computer vision, and GPS-based software systems; skilled in building functional prototypes and solving technical problems across hardware, software, and manufacturing.

## Education

### B.S. in Mechatronics Engineering (In progress)

*Expected Dec. 2026*

Universidad Iberoamericana Puebla

- **GPA:** 9.1/10.0. **Relevant coursework:** Strength of Materials, Advanced Control Systems, Embedded Systems, Mechanical Design, Industrial Automation.

### B.S. in Biomedical Engineering (Graduated with Honorable Mention)

*2020 - 2024*

Universidad Iberoamericana Puebla

- **GPA:** 9.1/10.0. Focus in biomechanics, medical device design, finite element analysis, and prototyping.

## Relevant Experience

### Technical Specialist & Medical Equipment Application Representative

*Jul. 2024 - Dec. 2024*

Punto Focal Equipo Médico, Puebla, Mexico

- Performed preventive/corrective maintenance, installation, and troubleshooting of ultrasound devices, X-ray systems, flat panels, and triggers.
- Supported 15+ ultrasound-related service cases, preparing technical reports, client follow-up, warranty/ticket documentation, and basic diagnostics.
- Delivered technical training and product demonstrations to physicians, technicians, and customers.
- Identified a functional replacement for a medical power supply, reducing client cost from approx. MXN \$15,000 to MXN \$1,500.

## Selected Projects

### 3-DOF Ball-Balancing Platform

*Mechatronics*

ESP32, SG90 servos, Python, OpenCV, PID control, Bluetooth communication

- Built a functional closed-loop mechatronic platform that uses camera-based feedback and continuous PID control to balance a ball at the center of the plate.
- Integrated embedded control, servo actuation, real-time vision processing, and PC-to-microcontroller communication for mechatronic prototyping.

### XGIO Smart Cane GPS Tracking Ecosystem

*Social Service*

Mobile app, backend, admin dashboard, Firebase, Vercel, CAD/SolidWorks documentation

- Developed a functional GPS-based system for people with visual impairment, including app, backend, administrative dashboard, user-device registration workflows, and technical documentation.

## Honors, Certifications and Skills

- **Honors:** Poster accepted at ISPO 20th World Congress 2025, Stockholm: "Redesign and validation of a tool holding system for a transradial prosthesis for a stomatology student"; Honorable Mention in Biomedical Engineering, Dec. 2024.
- **Certifications:** CSWA, Certified SOLIDWORKS Simulation Associate, Certified SOLIDWORKS Additive Manufacturing Associate, Google Project Management Certificate, EF SET English B2.
- **Technical skills:** SolidWorks, SolidWorks Simulation, MATLAB, Python, OpenCV, Arduino, ESP32, Firebase, Altium Designer, Power BI, Excel, CNC/router CNC, lathe, milling, laser cutting, 3D printing, drawing interpretation, and metrology.